FIITJEE

ALL INDIA TEST SERIES

OPEN TEST – II

JEE (Advanced)-2021

PAPER – 2

TEST DATE: 21-03-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A, C

Sol. $P = P_0 + \frac{4\sigma}{r}$

Since outside pressure, $P_0 = 0$

$P = \frac{4\sigma}{r} \Rightarrow PV^{1/3}$ = constant

Hence molar heat capacity of the gas, $C = C_V + \frac{R}{(1-x)} = \frac{5R}{2} + \frac{3R}{2}$

$\Rightarrow C = 4R$

Now, when the radius of the soap bubble is increased by dr = x(say) from equilibrium, then the net restoring force acting on the soap bubble

$F = 2 \times \frac{4\sigma}{r^2} dr 4\pi r^2 = 32\sigma\pi dr = 32\sigma\pi x$

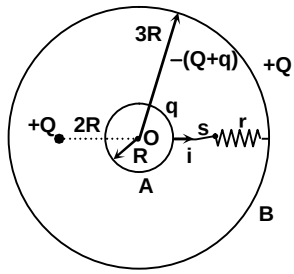
Now, $\rho 4\pi r^2 h \frac{dv}{dt} = -32\sigma\pi x$

$\frac{d^2x}{dt^2} = -\left(\frac{8\sigma}{\rho hr^2}\right)x$

Time period of small radial oscillations of the soap film, $T = 2\pi\sqrt{\frac{\rho hr^2}{8\sigma}}$

2. B, C, D

Sol. In steady state,



$V_A = V_B$

$$\frac{kq}{R}+\frac{kQ}{2R}-\frac{kq}{3R}=\frac{kQ}{3R}$$

$$\frac{2kq}{3R}=-\frac{kQ}{6R}$$

$$\Rightarrow q=-\frac{Q}{4}$$

At any time 't' the current through the resistor 'r' is

$$ir=\left(\frac{kq}{R}-\frac{kq}{3R}+\frac{kQ}{2R}\right)-\frac{kQ}{3R}$$

$$ir=\frac{2kq}{3R}+\frac{kQ}{6R}=\frac{k}{6R}(4q+Q)$$

$$i=\frac{k}{6Rr}(4q+Q)=\alpha(4q+Q) \qquad \left[\text{where } \alpha=\frac{k}{6Rr}\right]$$

$$-\frac{dq}{dt}=\alpha(4q+Q) \Rightarrow -\int_0^q\frac{dq}{(4q+Q)}=\alpha\int_0^t dt$$

$$\Rightarrow \ell n\left(\frac{4q+Q}{Q}\right)=-4\alpha t$$

$$q=\frac{Q}{4}\left[e^{-4\alpha t}-1\right]$$

$$i=-\frac{dq}{dt}=Q\alpha e^{-4\alpha t}$$

Now the total heat dissipated in the resistance 'r' is

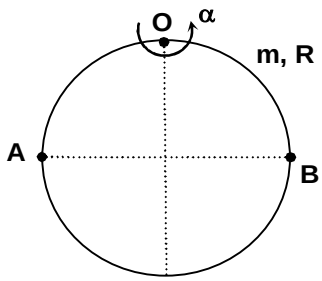
$$H=\int_0^\infty i^2 r dt=Q^2\alpha^2 r\int_0^\infty e^{-8\alpha t}dt=\frac{Q^2\alpha^2 r}{8\alpha}$$

$$H=\frac{Q^2\alpha r}{8}=\frac{Q^2 r}{8}\left(\frac{k}{6Rr}\right)=\frac{kQ^2}{48R}$$

$$H=\frac{Q^2}{192\pi\varepsilon_0 R}$$

3. A, D

Sol. $\tau = I\alpha$



$mgR - NR = 4mR^2\alpha$

$mg - N = 4m\alpha R$ …(i)

$N = m\alpha R$ …(ii)

From (i) and (ii)

$mg = 5m\alpha R$

$\alpha = g/5R$

$\therefore$ acceleration of bead 'A' is

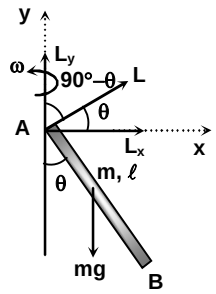
$$a_A=\alpha R\sqrt{2}=\frac{g\sqrt{2}}{5}=2\sqrt{2}\ m/s^2$$

$\therefore$ acceleration of bead 'B' is $a_B=\frac{g}{5}\sqrt{26}=2\sqrt{26}\ m/s^2$

Then normal force on the bead 'B' due to ring $N = m\alpha R = mg/5$

4. A, C

Sol. Total angular momentum of the rod about end 'A' is $L = \frac{m\ell^2}{3}\omega\sin\theta$



$L_x = L\cos\theta = \frac{m\ell^2}{3}\omega\sin\theta\cos\theta$

$L_y = L\sin\theta = \frac{m\ell^2}{3}\omega\sin^2\theta$

$\vec{L} = \vec{L}_x + \vec{L}_y$

$\frac{d\vec{L}}{dt} = \frac{d\vec{L}_x}{dt} + \frac{d\vec{L}_y}{dt}$

$\frac{d\vec{L}}{dt} = \frac{d\vec{L}_x}{dt}$

$\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d\vec{L}_x}{dt}$

$mg\frac{\ell}{2}\sin\theta = L_x\omega$

$mg\frac{\ell}{2}\sin\theta = \frac{m\ell^2}{3}\omega^2\sin\theta\cos\theta$

$\omega = \sqrt{\frac{3g}{2\ell\cos\theta}} = \sqrt{\frac{3\times10\times5}{2\times1\times3}} = 5$ rad/s

Now let F_h and F_v are the horizontal and vertical components of hinge reaction acting on the rod at end 'A'

$F_h = \frac{m\omega^2\ell}{2}\sin\theta = 1\times25\times\frac{1}{2}\times\frac{4}{5} = 10$ N

$F_V = mg = 10$ N

The net hinge reaction

$F = \sqrt{F_h^2 + F_v^2} = 10\sqrt{2}$ N

5. B, C

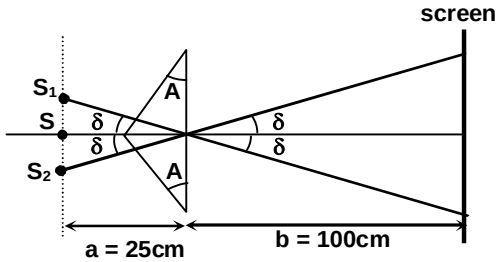
Sol. $\delta = (\mu - 1)A = (1.5 - 1)4 = 2 = 2\times\frac{\pi}{180} = \frac{\pi}{90}$ radian

The separation between two images formed is $S_1S_2 = 2a\delta$

$d = 2a\delta$

and $D = (a + b)$

$\therefore$ fringe width, $\omega = \frac{\lambda D}{d} = \frac{\lambda(a+b)}{2a\delta} = \frac{6\times10^{-7}}{2\times\frac{\pi}{90}}\left(\frac{125}{25}\right) = 43\,\mu m$



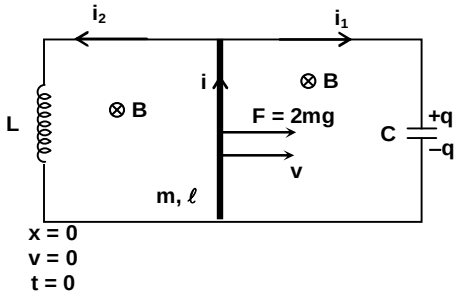
If the source is located at large distance form the biprism ($a \to \infty$)

$\therefore$ fringe width, $\omega = \frac{\lambda}{2\delta}\left(\frac{a+b}{a}\right) = \frac{\lambda}{2\delta}\left(1+\frac{b}{a}\right) = \frac{\lambda}{2\delta}$

$$\omega = \frac{\lambda}{2\delta} = \frac{6\times10^{-7}}{2\times\frac{\pi}{90}} = 8.6\ \mu m$$

6. B, D

Sol. $\varepsilon = Bv\ell$



$q = C\varepsilon = CBv\ell$

$$i_1 = \frac{dq}{dt} = CB\ell\frac{dv}{dt} \qquad \ldots(i)$$

$$\frac{Ldi_2}{dt} = Bv\ell$$

$$\int di_2 = \left(\frac{B\ell}{L}\right)\int v dt$$

$$\int_0^{i_2} di_2 = \frac{B\ell}{L}\int_0^x dx$$

$$i_2 = \left(\frac{B\ell}{L}\right)x \qquad \ldots(ii)$$

$$m\frac{dv}{dt} = F - Bi\ell$$

$$\frac{mdv}{dt} = F - B\ell(i_1 + i_2)$$

$$\frac{mdv}{dt} = F - B\ell i_1 - B\ell i_2$$

$$m\frac{dv}{dt} = 2mg - B\ell\left(CB\ell\frac{dv}{dt}\right) - B\ell\left(\frac{B\ell}{L}\right)x$$

$$(m + CB^2\ell^2)\frac{d^2x}{dt^2} = -\frac{B^2\ell^2}{L}\left(x - \frac{2mgL}{B^2\ell^2}\right)$$

$$\frac{d^2x}{dt^2} = -\frac{B^2\ell^2}{2mL}\left(x - \frac{2mgL}{B^2\ell^2}\right) \quad [\because CB^2\ell^2 = m)$$

$\therefore$ angular frequency of SHM executed by the rod is

$$\omega = \frac{B\ell}{\sqrt{2mL}}$$

The amplitude of SHM executed by the rod is

$$A = \frac{2mgL}{B^2\ell^2}$$

The maximum horizontal displacement of the rod

$$x_{max} = 2A = \frac{4mgL}{B^2\ell^2}$$

7. A, C

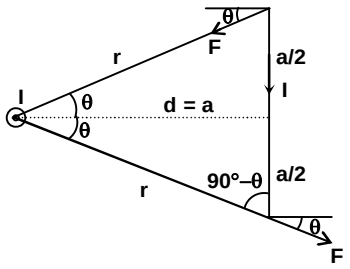
Sol. The net force acting on the square loop,

$F_0 = 2F \sin\theta$

$= 2BIa \sin\theta$

$= \frac{2\mu_0 I^2 a}{2\pi r}\frac{a}{2r}$

$F_0 = \frac{2\mu_0 I^2}{5\pi}$



The torque experienced by the square loop,

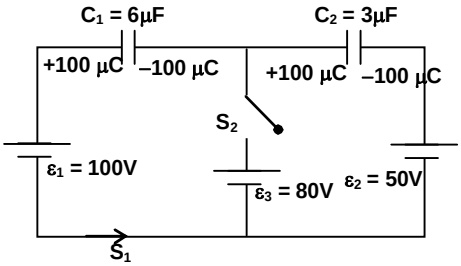
$\tau = BIa\cos\theta\, a = BIa^2\cos\theta$

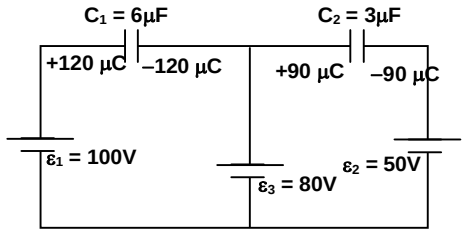
$\tau = \frac{\mu_0 I^2 a^2}{2\pi r}\frac{a}{r} = \frac{2\mu_0 I^2 a}{5\pi}$

$\tau = \frac{2\mu_0 I^2 a}{5\pi}$

8. B, C, D

Sol.





The charge flown through the battery ε_1 and ε_2 after closing the switch 'S_2',

$\Delta Q_1 = 120 - 100 = 20\ \mu C$

$\Delta Q_2 = -90 - (-100) = 10\ \mu C$

Now, $\Delta W_b = \Delta U + \Delta H$

$20\times10^{-6}\times100 + 10\times10^{-6}\times50 - 30\times10^{-6}\times80 =$

$$\frac{1}{2}\times6\times10^{-6}\times400 + \frac{1}{2}\times3\times10^{-6}\times900 - \frac{1}{2}\times2\times10^{-6}\times2500 + \Delta H$$

$\Rightarrow 100\times10^{-6} = 50\times10^{-6} + \Delta H$

$\Rightarrow \Delta H = 50\times10^{-6} = 50\ \mu J$

9. A

Sol. The magnetic pressure exerted on the wall of inner cylinder.

(I) $P_i = \frac{\mu_0}{2}\left(\frac{I}{2\pi a}\right)^2 = \frac{\mu_0 I^2}{8\pi^2 a^2}$

(II) $P_i = \frac{\mu_0}{2}\left(\frac{2I}{2\pi a}\right)^2 = \frac{\mu_0 I^2}{2\pi^2 a^2}$

(III) $P_i = \frac{\mu_0}{2}\left(\frac{I}{2\times2\pi a}\right)^2 = \frac{\mu_0 I^2}{32\pi^2 a^2}$

(IV) $P_i = \frac{\mu_0}{2}\left(\frac{3I}{2\times2\pi a}\right)^2 = \frac{9\mu_0 I^2}{32\pi^2 a^2}$

10. C

Sol. The magnetic pressure exerted on the wall of outer cylinder.

(I) $P_0 = \frac{I}{4\pi a}\left(\frac{\mu_0 I}{4\pi a} - \frac{\mu_0 I}{8\pi a}\right) = \frac{\mu_0 I^2}{32\pi^2 a^2}$

(II) $P_0 = \frac{2I}{4\pi a}\left(\frac{\mu_0 2I}{4\pi a} - \frac{\mu_0 2I}{8\pi a}\right) = \frac{\mu_0 I^2}{8\pi^2 a^2}$

(III) $P_0 = \frac{2I}{4\pi a}\left(\frac{\mu_0 I}{8\pi a} + \frac{\mu_0 2I}{8\pi a}\right) = \frac{3\mu_0 I^2}{16\pi^2 a^2}$

(IV) $P_0 = \frac{3I}{4\pi a}\left(\frac{\mu_0 3I}{8\pi a} + \frac{\mu_0 3I}{8\pi a}\right) = \frac{9\mu_0 I^2}{16\pi^2 a^2}$

11. B

12. C

Sol. **(For Q.11 to Q.12):**

(I) $\Delta W_{cycle} = \frac{1}{2} \times V_0 \times P_0 = \frac{P_0 V_0}{2}$

$\Delta Q_{AB} = nC\Delta T = n2R(4T_0 - T_0) = 6nRT_0 = 6P_0V_0$

$\Delta Q_{BC} = nC_V\Delta T = n\frac{3R}{2}(2T_0 - 4T_0) = -3nRT_0 = -3P_0V_0$

$\Delta Q_{CA} = nC_P\Delta T = \frac{5}{2}nR(T_0 - 2T_0) = -\frac{5}{2}nRT_0 = -\frac{5}{2}P_0V_0$

Total heat absorbed by the gas = $6P_0V_0$

(II) $\Delta Q_{AB} = nC_P\Delta T = \frac{5}{2}nR\Delta T = \frac{5}{2}P_0V_0$

$\Delta Q_{BC} = nC\Delta T = n\frac{R}{2}(T_0 - 2T_0) = -\frac{nRT_0}{2} = -\frac{P_0V_0}{2}$

$\Delta Q_{CA} = nRT_0\ell n\left(\frac{V_0}{4V_0}\right) = -2nRT_0\ell n2 = -2P_0V_0 \ln 2$

$\Delta W_{cycle} = \frac{5P_0V_0}{2} - \frac{P_0V_0}{2} - 2P_0V_0\ell n2 = 2P_0V_0(1 - \ell n2)$

Total heat absorbed by the gas = $\frac{5P_0V_0}{2}$

(III) $\Delta Q_{AB} = nC\Delta T = n\frac{3R}{2}\left(\frac{T_0}{2} - T_0\right) = -\frac{3}{4}nRT_0 = -\frac{3}{4} \times 4P_0V_0 = -3P_0V_0$

$\Delta Q_{BC} = nC_V\Delta T = n\frac{5R}{2}\left(\frac{T_0}{4} - \frac{T_0}{2}\right) = -\frac{5}{2}\frac{nRT_0}{4} = -\frac{5P_0V_0}{2}$

$\Delta Q_{CA} = nC\Delta T = n2R\Delta T = 2nR\left(T_0 - \frac{T_0}{4}\right) = \frac{6nRT_0}{4} = 6P_0V_0$

$\Delta W_{cycle} = \Delta Q_{cycle} = 6P_0V_0 - 3P_0V_0 - \frac{5P_0V_0}{2} = \frac{P_0V_0}{2}$

Total heat absorbed by the gas = $6P_0V_0$

(IV) $\Delta Q_{AB} = nC_P\Delta T = \frac{7}{2}nR\Delta T = \frac{7}{2}P_0V_0$

$\Delta Q_{BC} = nR2T_0\ell n2 = 2P_0V_0\ell n2$

$\Delta Q_{CD} = nC_V\Delta T = n\frac{5R}{2}(T_0 - 2T_0) = -\frac{5}{2}nRT_0 = -\frac{5}{2}P_0V_0$

$\Delta Q_{DA} = nRT_0\ell n\left(\frac{1}{4}\right) = -2nRT_0\ell n2 = -2P_0V_0\ell n2$

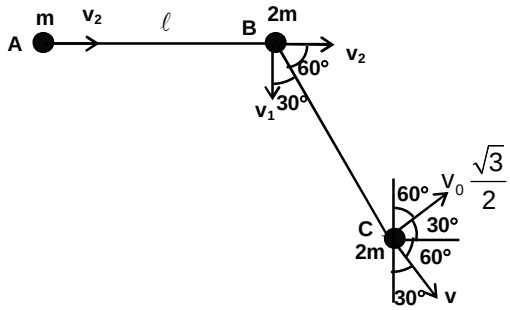
$\Delta W_{cycle} = \Delta Q_{cycle} = \frac{7P_0V_0}{2} - \frac{5P_0V_0}{2} = P_0V_0$

Total heat absorbed by the gas = $\frac{7P_0V_0}{2} + 2P_0V_0\ell n2 = \frac{P_0V_0}{2}(7 + 4\ell n2)$

SECTION – C

13. 00001.80

Sol. Using conservation of momentum of system.



$2mv_0 = 3mv_2 + 2m\left(\frac{3v_0}{4} + \frac{v}{2}\right)$

$2v_0 = 3v_2 + v + \frac{3v_0}{2}$

$3v_2 + v = \frac{v_0}{2}$ …(i)

$2mv_1 = 2m\left(\frac{\sqrt{3}v_0}{4} - \frac{v\sqrt{3}}{2}\right)$

$v_1 = \frac{\sqrt{3}v_0}{4} - \frac{\sqrt{3}v}{2}$

$\sqrt{3}v_1 = \frac{3v_0}{4} - \frac{3v}{2}$ …(ii)

Also, $v_1\cos30° + v_2\cos60° = v$

$\frac{\sqrt{3}v_1}{2} + \frac{v_2}{2} = v$

$\sqrt{3}v_1 = 2v - v_2$ …(iii)

From equation (ii) and (iii)

$2v - v_2 = \frac{3v_0}{4} - \frac{3v}{2}$

$7v - 2v_2 = \frac{3v_0}{2}$ ….(iv)

Solving equation (i) and (iv)

$23v_2 = 2v_0$

$v_2 = \frac{2v_0}{23} = \frac{2\times20.7}{23} = 1.8$ m/s

14. 00037.50

Sol. $a_1 \cos\theta = a_2 \cos\theta$

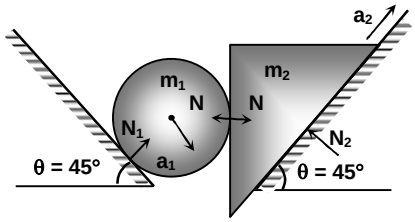
$a_1 = a_2 = a$ (say)

$m_1 g \sin\theta - N\cos\theta = m_1 a$ …(i)

$N\cos\theta - m_2 g \sin\theta = m_2 a$ …(ii)

Solving (i) and (ii) we get

$$a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g\sin\theta \quad \text{…(iii)}$$



From equation (i) and (iii)

$N\cos\theta = m_1 g\sin\theta - m_1 g\sin\theta$

$$\left(\frac{m_1 - m_2}{m_1 + m_2}\right)$$

$$N\cos\theta = \frac{2m_1 m_2 g\sin\theta}{(m_1 + m_2)}$$

$$N = \frac{2m_1 m_2 g\tan\theta}{(m_1 + m_2)} = \frac{2\times 5\times 3\times 10\times 1}{8} = 37.5 \text{ N}$$

N = 37.50 N

15. 00000.60

Sol. Total mechanical energy

$$\varepsilon = \frac{-GMm}{r_0} + \frac{1}{2}mv_0^2$$

$$= -\frac{GMm}{r_0} + \frac{1}{2}mv_0^2$$

$$= -\frac{GMm}{r_0} + \frac{GMm}{2r_0}$$

$$\varepsilon = -\frac{GMm}{2r_0}$$

$$\frac{-GMm}{2a} = -\frac{GMm}{2r_0}$$

$\therefore$ Length of semi-major axis, $a = r_0$

Hence point 'P' must be an extreme

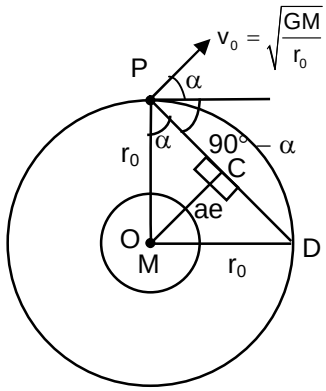
Point of the minor axis of the elliptical orbit

$\because$ $a\sin\alpha = ae$

$e = \sin\alpha = \sin 37° = 0.6$

Hence the eccentricity of the elliptical orbit

e = 0.60



16. 00020.00

Sol. The tension produced in each wire,

$$F = \frac{(\alpha_1 + \alpha_2)\ell\Delta\theta}{\frac{\ell}{A}\left(\frac{1}{Y_1} + \frac{1}{Y_2}\right)} = \frac{3\times 10^{-5}\times 20\times 10^{-5}}{1.5\times 10^{-10}} = 40\text{N}$$

Now, $\frac{n_1}{2\ell}\sqrt{\frac{F}{\mu_1}} = \frac{n_2}{2\ell}\sqrt{\frac{F}{\mu_2}}$

$\Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{\mu_1}{\mu_2}} = \sqrt{\frac{0.4}{0.1}} = 2$

In first overtone, $n_1 = 4$, $n_2 = 2$

Hence the first overtone frequency of the transverse vibration of the system,

$f = \frac{n_1}{2\ell}\sqrt{\frac{F}{\mu_1}}$

$f = \frac{4}{2\times 1}\sqrt{\frac{40}{0.4}} = \frac{4}{2}\times 10 = 20\ \text{Hz}$

17. 00001.84

Sol. $E_1 = k_1 + \phi$ …(i)

$E_2 = k_2 + \phi$ …(ii)

Also, $\frac{k_1}{k_2} = 4 \Rightarrow k_1 = 4k_2$ …(iii)

From (i) and (ii)

$\frac{E_1}{E_2} = \frac{k_1+\phi}{k_2+\phi}$

$\frac{\lambda_2}{\lambda_1} = \frac{4k_2+\phi}{k_2+\phi}$

$\frac{6000}{4500} = \frac{4k_2+\phi}{k_2+\phi}$

$\frac{4}{3} = \frac{4k_2+\phi}{k_2+\phi}$

$\Rightarrow 4k_2 + 4\phi = 12k_2 + 3\phi$

$\Rightarrow \phi = 8k_2$ …(iv)

Now form (ii) and (iv)

$E_2 = k_2 + 8k_2$

$\Rightarrow \frac{hC}{\lambda_2} = 9k_2$

$\Rightarrow \frac{1242}{600} = 9k_2$

$\Rightarrow 2.07 = 9k_2$

$\Rightarrow k_2 = 0.23$ eV

$\therefore \phi = 8k_2 = 8 \times 0.23 = 1.84$ eV

18. 00006.82

Sol. The pitch of screw gauge, P = 1mm

Least count, $\text{L.C.} = \frac{P}{N} = \frac{1}{100} = 0.01\,\text{mm}$

Diameter of the wire,

$d = 2\text{mm} + 56 \times \text{L.C.} - 6 \times \text{L.C.}$

$= 2\text{mm} + 56 \times 0.01\ \text{mm} - 6 \times 0.01\ \text{mm}$

$d = 2.50\ \text{mm} = 0.250\ \text{cm}$

$\ell = 8.68$ cm

Curved surface area of the wire,

$S = \pi d\ell$

$= \frac{22}{7}\times 0.250 \times 8.68 = 6.82\ \text{cm}^2$

Chemistry

PART – II

SECTION – A

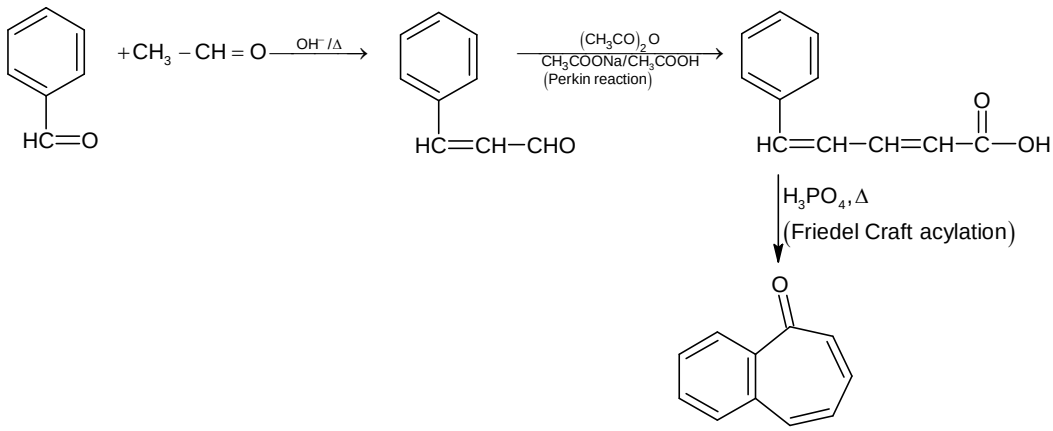
19. A, B, C

Sol. Average C – O bond length in metal compounds $\propto \frac{\text{No. of } d\pi - \text{electron pairs}}{\text{No. of CO groups}}$

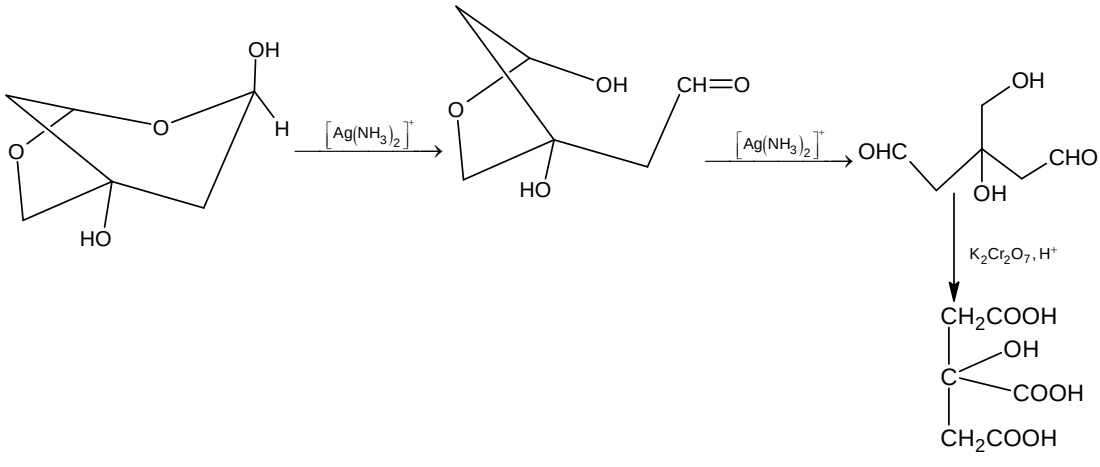
Complex	Ratio
$[Fe(CO)_5]$	0.8
$[Cr(CO)_6]$	0.5
$[Cr(NH_3)(CO)_5]$	0.6
$[Cr(NH_3)_2(CO)_4]$	0.75
$[Cr(NH_3)(CO)_3]$	1.0

20. A, B, C, D

Sol. (A)

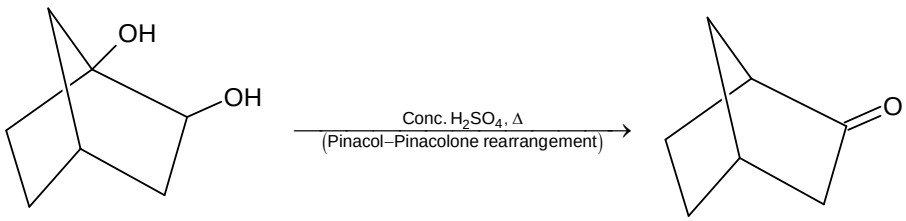


(B)

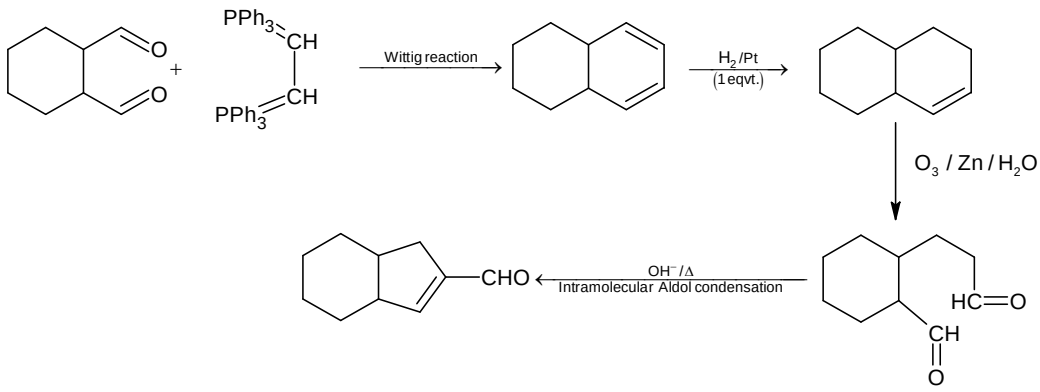


Citric acid

(C)



(D)



21. A, B, D

Sol.

$$2ACl_3 \longrightarrow A_2O_5$$

at $t = 0$ 1M 0

After time $= 10$ min $(1-x)$ $\frac{x}{2}$

So, $(1-x)\times 2 = \frac{1}{4}\times 0.8\times 5$

$\therefore x = 0.5$ mole

So, $t_{1/2} = 10\,\text{min}$

Now, $k = \frac{1}{2\times t_{1/2}\times [A]_o} = \frac{1}{2\times 10\times 1} = 0.05\ \ell\,\text{mol}^{-1}\,\text{min}^{-1}$

Also, after t = 20 min,

$$20 = \frac{1}{2\times 0.05}\left[\frac{1}{[A]_t} - \frac{1}{[A]_o}\right]$$

$$2 = \frac{1}{[A]_t} - 1$$

$$\therefore [A]_t = \frac{1}{3}$$

So, % of ACl_3 left unreacted after 20 min $= \frac{1}{3}\times 100 = 33.33\%$

22. A, B, C

Sol. (i)

$$PO_4^{3-} + H^+ \longrightarrow HPO_4^{2-}$$

m.mol.	10	5	0
after neutralization	5	0	5

So, $pH = pK_{a_3} = 13$ (Buffer)

(ii)

$$PO_4^{3-} + H^+ \longrightarrow HPO_4^{2-}$$

m.mol.	10	10	0
after neutralization	0	0	10

So, $pH = \frac{8+13}{2} = 10.5$ (Amphiprotic anion)

(iii)

	PO_4^{3-} +	$H^+ \longrightarrow$	HPO_4^{2-}
m.mol.	10	15	0
after neutralization	0	5	10

HPO_4^{2-} +	$H^+ \longrightarrow$	$H_2PO_4^-$
10	5	0
5	0	5

So, $pH = pK_{a_2} = 8$ (Buffer)

(iv)

	PO_4^{3-} +	$H^+ \longrightarrow$	HPO_4^{2-}
m.mol.	10	20	0
after neutralization	0	10	10

	HPO_4^{2-} +	$H^+ \longrightarrow$	$H_2PO_4^-$
m.mol.	10	10	0
after neutralization	0	0	10

So, $pH = pK_{a_3} = 5.5$ (Amphiprotic anion)

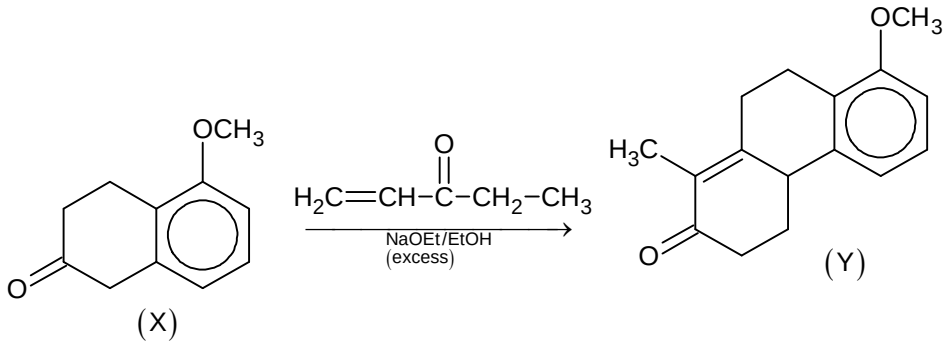
23. A, B, C

Sol. Mixing of gases is always accompanied with the increase in the entropy.

- At equilibrium, chemical potential is same in all phases of the system.
- Since in reversible adiabatic process, dS = 0, therefore these are isoentropic process.
- During expansion, work is maximum under reversible conditions.

24. B, C, D

Sol.

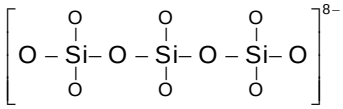


25. A, D

Sol. The order of volatility is IV < III < II < I

So, I = CH_3F, II = CH_2O, III = CH_3OH, IV = CH_3COOH

26. A, B, C

Sol. (A) 

(B) HF reacts with glass, so it is used to make marking on the glass (etching).

(C) Since Fe^{3+} reacts with KCNS to produce red colour. So, it can be used as an indicator in the titration of Fe^{3+} with Sn^{2+}.

(D) $Pb_3O_4 + HCl \longrightarrow PbCl_2 + H_2O + \underset{\text{(Greenish yellow)}}{Cl_2} \uparrow$

27. D

Sol. (A) Spontaneity requires ΔG to be negative. Here ΔH is positive but nothing can be said about ΔS, so the process may or may not be spontaneous. Hence, option (A) is incorrect.

(B) Since, $\Delta S = 0$ so $\Delta G = \Delta H$ but $\Delta H = \Delta U + P\Delta V$, so this process can be spontaneous if $\Delta V$is negative or zero or even small positive. But nothing is mentioned about ΔV, so the process may or may not be spontaneous. So, (B) is incorrect.

(C) For a reversible adiabatic process,
$\Delta S_{Syst} = \Delta S_{surr} = 0$ and for an irreversible adiabatic process, $\Delta S_{syst} > 0$ and $\Delta S_{surr} = 0$
So, (C) is incorrect.

28. B

Sol. An adiabatic process implies q = 0, and with volume not changing we have $\Delta V = 0$; therefore, W = 0 and thus $\Delta U = 0$. The constant 'U' and 'V' allows us to apply the strict entropy spontaneity test; if $\Delta S > 0$ the process is spontaneous. Since, we are given that $\Delta S$is +ve, so the process must be spontaneous.

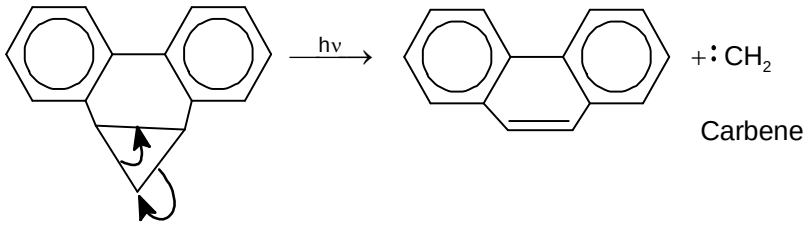
$\Delta G = \Delta H - T\Delta S$

Here, $\Delta H = \Delta U = 0$, $\Delta S = +ve$

So, ΔG must be negative.

29. B

Sol.



30. D

Sol. (A) Carbene is a diradical.
(B) It is Favorskii Rearrangement, where carbanion is formed as intermediate.
(C) It is Beckman's Rearrangement, where carbocation is the intermediate.

SECTION – C

31. 00006.50

Sol. All the above mentioned compounds except N_2 can react with PCl_5 to give $POCl_3$.

32. 00005.30

Sol. $\underset{(1\,bar)}{Pt;H_2(g)} \,|\, HA \,||\, AgCl(s), Cl^- \,|\, Ag(s)$

The cell reaction is

$$\frac{1}{2}H_2(g) + AgCl(s) \longrightarrow H^+ + Cl^- + Ag(s)$$

$E_{cell} = E^\circ_{cell} - 0.06\log_{10}\left[H^+\right]\left[Cl^-\right]$

$= 0.20 - 0.06 \times \log\left[Cl^-\right] + 0.06 \times pH$

$E_{cell} = 0.26 + 0.06 \times pH$ ... (1)

$0.56 = 0.26 + 0.06 \times pH_1$

$pH_1 = 5$

Similarly: $0.62 = 0.26 + 0.06 \times pH_2$

$pH_2 = 6$

Now, let there be initially x mol HA present and 20 mL NaOH containing 'y' mol of NaOH.

So, $pH_1 = 5 = pK_a + \log\frac{y}{x-y}$ … (2)

$pH_2 = 6 = pK_a + \log\frac{2.5y}{x-2.5y}$ … (3)

On solving Eqn. (1) and (2), we get

$x = 3y$

Now, substituting x = 3y in Eqn. (2)

We get $pK_a = 5.30$

33. 00000.75

Sol. I, II, V and VII will exist as anion at pH = 7, So, y = 4 and rest III, IV and VI will exist as cation at pH = 7. So, x = 3.

So, $\frac{x}{y} = 0.75$

34. 00004.50

Sol. x = 4 (I, II, III, VII)
y = 5 (I, II, III, IV, VII)
z = 2 (V, VI)

35. 00025.76

Sol. Meqvt. of Fe^{2+} taken = 50 × 0.16 = 8

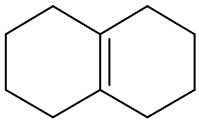
- Meqvts. of Fe^{2+} ion left = 3.67 × 0.05 × 6 = 1.10
- Meqvts. of Cr^{+6} in solution = 8 – 1.10 = 6.90
- m. mol. of Cr^{+6} in solution = 6.90/3 = 2.30
- m. mol. of Chromite = 2.30/2 = 1.15

Mass of chromite in original sample $= \frac{1.15 \times 224}{1000} = 0.2576$ g

% of chromite in the original sample = 25.76%

36. 00001.20

Sol.

X = 

So, x = 3, y = 5.

Mathematics

PART – III

SECTION – A

37. A, B, C, D

Sol. $\frac{\left(1-\frac{b}{a}\right)\left(2-\frac{b}{a}\right)}{\left(1-\frac{b}{a}+\frac{c}{a}\right)}=\frac{\alpha}{\beta+1}+\frac{\beta}{\alpha+1}+2$. Let $0\le\alpha\le\beta\le1\Rightarrow\frac{\alpha}{\beta+1}+\frac{\beta}{\alpha+1}\le\frac{\alpha}{\alpha+1}+\frac{1}{\alpha+1}\le1$

Maximum value is 3, if $\alpha = 0$, $\beta = 1$ or $\alpha = 1$, $\beta = 1$ and minimum value = 2, if $\alpha = \beta = 0$

38. A, C, D

Sol. $a_{n+1}^2-a_n^2=1$. By Telescopic sum $a_n=\sqrt{n-1+a_1^2}$ now $a_{2n_0}=3a_{n_0}\Rightarrow 8a_1^2=8-7n_0\ge0\Rightarrow n_0=1$

$\Rightarrow a_1^2=\frac{1}{8}\Rightarrow a_n=\sqrt{n-\frac{7}{8}}$ and $\sqrt{\frac{8}{8a_n^2+7}}=\frac{1}{\sqrt{n}}$, $\sqrt{n+1}-\sqrt{n}<\frac{1}{2\sqrt{n}}<\sqrt{n}-\sqrt{n-1}$

$\Rightarrow \sqrt{50}-1<\frac{1}{2}\sum_{n=1}^{49}\sqrt{\frac{8}{8a_n^2+7}}<7$

39. A, B, D

Sol. (A) Number of ways so that the sum of a common edge is odd = $2\times4!\times4!$

(B) Number ways assigning numbers such that the sum of any two numbers with a common body diagonal even is $6\times(4!)^2$

40. A, D

Sol. $g(a,b)=\frac{a^2+b^2+ab}{\sqrt{3}}$

41. C, D

Sol. Total number of points will be 17 and number of squares joining these 16 point will be equal to $(3^2\times1+2^2\times2+1^2\times3)=20$

42. A, B

Sol. After 5 second bug can reach at (5, 0), (4, 1), (3, 2), (2, 3), (1, 4) and (0, 5) along 1, 5, 10, 10, 5, 1 different paths.

43. B, D

Sol. $r_1=s\tan\left(\frac{A}{2}\right)\Rightarrow\frac{r_1}{s}=\tan\left(\frac{A}{2}\right)$; $a>r_1\Rightarrow1>\frac{a}{s}>\tan\left(\frac{A}{2}\right)\Rightarrow$ A is acute and $r=(s-a)\tan\left(\frac{A}{2}\right)$

44. B, D

Sol. The point on S for which are of triangle with point B and C will be minimum, is $\left(\frac{5}{2},\frac{25}{4}\right)$

So, minimum area = $\frac{15}{4}$ unit^2 and $\sqrt{(\alpha-2)^2+\beta^2}=\sqrt{\left(\frac{5}{2}-2\right)^2+\left(\frac{25}{4}\right)^2}=\frac{\sqrt{629}}{4}$

45. C

46. B

Sol. **(for Q.45 to Q.46):**

$P \in B$, $Q \in A$, $R \in D$, $S \in C$ and $T \in A, C, D$; $U \in A, C, D$

47. C

48. D

Sol. **(for Q.47 to 48):**

$$\lim_{x\to 0}\frac{f(x)}{g(x)}=1;\ \lim_{x\to\infty}\frac{f(x)}{g(x)}=\text{D.N.E.}\ \Rightarrow\ \lim_{x\to 0}\frac{f'(x)}{g'(x)}=1;\ \lim_{x\to\infty}\frac{f'(x)}{g'(x)}=0$$

$$\lim_{x\to 0}\frac{h(x)}{t(x)}=0;\ \lim_{x\to\infty}\frac{h(x)}{t(x)}=0\ \Rightarrow\ \lim_{x\to 0}\frac{h'(x)}{t'(x)}=\text{D.N.E.};\ \lim_{x\to\infty}\frac{h'(x)}{t'(x)}=0$$

SECTION – C

49. 00002.33

Sol. If both go to same bin, then probability = $\frac{1}{4}+\frac{1}{4^2}+\frac{1}{4^3}.....\infty=\frac{1}{3}$

So, required probability $(P)=\frac{1}{2}\left(1-\frac{1}{3}\right)=\frac{1}{3}$

50. 00007.00

Sol. Let $z = (1 + ai)(1 + bi)(1 + ci) \Rightarrow |z| = 3$

$|Re(z)| \le |z| \Rightarrow -2 \le ab + bc + ca \le 4$

51. 00039.00

Sol. The integral will be at its maximum value if $380 - x - x^2 \ge 0$ $x \in [-20, 19]$

$\Rightarrow a = 20, b = 19$

52. 00004.00

Sol. $f(x)=-\frac{x}{2}$, $\forall\ x \in R$, $f(x)=-\frac{x}{3}$, $\forall\ x \in R$

$$f(x)=\begin{cases}-\frac{x}{2} & ;\ x\ge 0\\ -\frac{x}{3} & ;\ x<0\end{cases},\quad f(x)=\begin{cases}-\frac{x}{3} & ;\ x\ge 0\\ -\frac{x}{2} & ;\ x<0\end{cases}$$

53. 00001.25

Sol. The exhaustive solution is $\left[0, \frac{\sqrt{5}-1}{2}\right)$

54. 02018.00

Sol. $f(1) = 1 \Rightarrow f(x)+f\left(\frac{2020}{x}\right)=2f(x) \Rightarrow f^2(x) = 1$

$$f(x)=\begin{cases}1 & ;\ x>0\\ -1 & ;\ x<0\end{cases}$$